

NETRĀ

The AI Growth Copilot for Every Paytm Merchant

● Track 1 — Merchant Growth AI

Team **KERNEL** · Utkarsh Singh (Team Leader) & Pranav Pant

India's Merchants Are Flying Blind

Ramesh runs a kirana store in Lajpat Nagar, Delhi. He has been a Paytm merchant for 4 years. He processes 80+ UPI transactions every day through his Soundbox. His monthly revenue is around ₹3.5 lakh. By every standard, he is a successful small merchant. But Ramesh has a problem he does not even know he has: *he cannot see past his own counter*. He does not know what products are trending two streets away. He does not know his snack prices are 14% above every other kirana around him. He does not know that last Navratri, he ran out of his top 3 items by day 4 because he ordered based on memory, not data. Ramesh is not failing. He is leaving growth on the table because he is running his business in the dark. And there are **4.9 crore merchants just like him on Paytm**.



No Market Visibility

A new chai brand is selling well everywhere around him. A competitor started selling namkeen cheaper. Ramesh finds out only when customers stop showing up. Zero outside window.



No Demand Signals

Navratri is in 3 weeks. Ramesh stocks based on gut feeling and last year's memories. He overstocks some items and runs out of others as every seasonal festival catches him off guard.

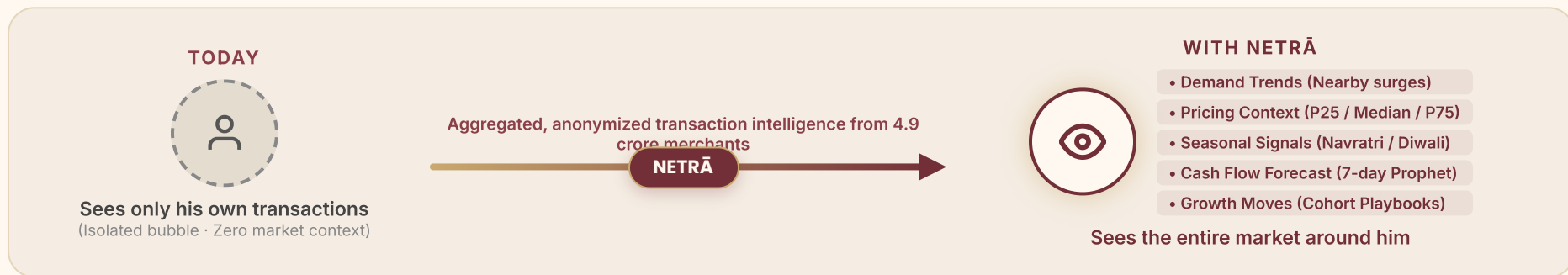


No Growth Playbook

Nobody tells Ramesh that peer kiranas adding weekend UPI offers saw 18% higher footfall. He learns by trial and error alone, while guidance data already exists inside Paytm.

Paytm processes the data of 4.9 crore merchants. The intelligence to solve all of this already exists. It just has not reached Ramesh yet.

What If Every Merchant Could See What the Network Sees?



Paytm already has the largest real-time commerce dataset in India. Every transaction, every merchant, every pin code, every season. **Netrā does not create new data.** It unlocks the intelligence that is already trapped inside Paytm's ecosystem and delivers it to each merchant as personalized, actionable growth moves. No merchant's individual data is ever exposed. *The network's patterns are the power.*

"Solo merchant. Network intelligence."

The Intelligence Layer — Reading the Market



Trade Radar

Hyperlocal Demand Heatmap

"See what nearby consumers are buying that you do not stock."

Netrā clusters every merchant into a hyperlocal micro-market — roughly a 1 to 2 km radius zone aligned with a real neighborhood (Lajpat Nagar, not "Delhi"). Within that zone, it aggregates transaction volumes by product category across all merchants. Beverages up 40% this week. Snacks steady. Personal care down 10%. Then it compares this market-level demand heatmap against YOUR transaction categories. If iced beverages are surging in Lajpat Nagar and Ramesh has zero transactions in that category, that is a gap. That gap becomes a Trade Radar alert: *"Iced tea is up 40% in your pin code. You stock none. Top 3 items to try based on what similar stores sell: [X, Y, Z]."* The ML underneath is collaborative filtering on a merchant-category matrix, localized by micro-market. Lightweight. Interpretable. Runs as a daily batch job on Paytm's existing transaction data.



Price Pulse

Statistical ATV Analysis

"Know where your pricing stands in the local market."

Netrā computes the average transaction value per product category per micro-market. Across all kiranas in Lajpat Nagar, the median snack transaction is ₹35. Ramesh's average is ₹40 — he is pricing 14% above the median. Netrā tracks this gap over time: is it widening? Are nearby merchants lowering prices? Is his volume dropping as a result? When it detects a risk, it does not just say "you are expensive" — that is demoralizing and useless. It frames it as an opportunity: *"Your snack pricing is 14% above the area average and your volume dipped 8% this week. A weekend combo — namkeen + cold drink at ₹30 — could pull price-sensitive customers back without a permanent price cut."* No merchant ever sees another merchant's prices. Only aggregate market averages. Privacy is absolute. The model is statistical outlier detection on average-transaction-value distributions, tracked over a rolling 30-day window.

The Action Layer — Moving Before the Market Does



Festival Engine

Auto-generates localized offers. One tap to send.

Netrā maintains a master calendar of Indian festivals — Diwali, Holi, Chhath Puja, Onam, Pongal, Baisakhi, and local melas. For each merchant, it pulls sales data from the same festival last year: what spiked, what sold out, what sat on shelves. Sarvam's LLM generates a complete, ready-to-send offer package: WhatsApp message in the merchant's language, inventory prep list (*"you ran out of X last time — order 20% more"*), and discount structures. One tap sends it via n8n's WhatsApp automation.



Cash Flow Prophet

Predicts next 7 days. Alerts before shortfalls hit.

Every merchant has a rhythm. Ramesh does ₹15,000 on Saturdays, ₹8,000 on Tuesdays, dips on Wednesdays. Netrā builds a daily revenue model from 90 days of transactions, capturing day-of-week rhythms, payday spending cycles (1st/15th), and seasonal baselines. If a shortfall looms, it recommends action: *"Thursday looks ₹4,000 short. Run a flash offer Tuesday to pull demand forward."* For Paytm, this doubles as a stealth underwriting signal for merchant credit.



Growth Missions

Weekly micro-tasks learned from peer merchants.

Netrā groups merchants into cohorts by location tier, category, and revenue. Within each cohort, it tracks which merchants GREW in the last 90 days and what preceded that growth. Netrā assigns a weekly mission: *"This week, add 3 items from the trending list. Stores like yours that did this saw 11% higher weekly revenue."* Outcome feedback loops back into Cognee to continuously strengthen peer signals. This is the flywheel. This is the moat.

Zero New Behavior. The Intelligence Comes to Them.



Netrā AI Engine Orchestration



Paytm Soundbox

Voice Briefing (Sarvam AI)

Every morning, the Soundbox speaks to the merchant in their language. A 30-second briefing: yesterday's numbers, nearby trends, offer results, and today's priority. Powered by Sarvam AI TTS in 11+ languages. No screens. They just listen while sweeping the store.

"Good morning Ramesh. Yesterday: 84 transactions, ₹12,400. Iced beverages are trending nearby. Today's focus: festival stock check."



Paytm for Business

App Intelligence Cards

A dedicated "Netrā" tab inside the Paytm for Business app. Zero new downloads. Interactive heatmaps for Trade Radar, 7-day cash flow forecast bars, and pre-packaged festival campaigns with 1-tap dispatch buttons for deeper exploration.

✓ Live inside the app they use daily



WhatsApp / SMS

Time-Sensitive Nudges (n8n)

For urgent signals — a festival 5 days away, a liquidity shortfall, or a sudden weekend heatwave spike — Netrā dispatches an automated WhatsApp nudge via n8n. The merchant simply replies "YES" to execute. Delivered right where they chat 50x daily.

✓ Reply 'YES' to launch offers

"Netrā does not ask merchants to adopt a new tool. It embeds intelligence into the tools they already trust."

System Architecture — Powered by Hackathon Partners

Paytm Transaction Data Stream

4.9 crore merchants · Billions of transactions · Location + Category + Amount + Timestamp

↓ ANONYMIZATION & DIFFERENTIAL PRIVACY AGGREGATION PIPELINE ↓

Merchant Knowledge Graph

Cognee

Persistent memory per merchant: historical trends, seasonal sales records, accepted suggestion feedback, native language preference, and cohort group assignments.

Netrā AI Engine

Core ML

- Geospatial Micro-clustering
- Category Heatmaps (Trade Radar)
- ATV Analysis (Price Pulse)
- 7-Day Prophet Forecasting
- Festival Decomposition
- Collaborative Growth Missions

Sarvam AI Indic LLM for contextual prompts in 11+ regional languages

Workflow Orchestration

n8n

Asynchronous event triggers: 6 AM Soundbox briefing scheduler, T-14 festival tender dispatch, nightly cash shortfall alarms, WhatsApp action callbacks, and ingestion pipelines.

● Soundbox Voice (Sarvam TTS)

● Paytm for Business App Cards

● WhatsApp / SMS Nudges (n8n)

↻ Continuous Feedback Loop: Merchant Actions & Conversions → Update Cognee Knowledge Graph → Auto-Fine-Tune Next Week's Missions

TECHNICAL STACK

Python 3.11 · FastAPI · Sarvam AI (LLM + Indic TTS) · Cognee (Graph Memory) · n8n (Automation) · PostgreSQL · React

Ramesh's First Week with Netrā

Four Proactive Moves. Seven Days. Measurable Compounding Growth.

MON MORNING

Soundbox

"Organic snacks are surging in your area. You stock none. Want 3 items from your wholesaler?"

→ Ramesh adds 2 items to next order

WED AFTERNOON

App Alert

"Your namkeen pricing is 14% above local average. A weekend combo at ₹30 could recover volume."

→ Ramesh enables combo in 1 tap

FRI EVENING

Soundbox

"Your combo brought 12 new customers today. Weekly revenue up ₹2,800. Cash flow looks healthy."

→ Ramesh trusts the next move

SUN NIGHT

Growth Mission

"Try buy-2-get-1 on your slowest namkeen. Peer stores saw 22% faster clearance with zero losses."

→ Ramesh queues offer for Monday

15 – 25%

Revenue Growth

Projected Q1 uplift per active merchant

4.9 Crore

Addressable Market

Paytm registered SME ecosystem base

30%+

Churn Reduction

Soundbox transformed into vital copilot

Why This Matters for Paytm



Subscription Stickiness

At ₹100–300/month, Soundbox is kept only while useful. Netrā elevates it from a passive payment chime into a vital daily intelligence asset. Merchants will never cancel, protecting Paytm's **₹1,500+ Cr** recurring subscription revenue.



Compounding Data Moat

Every active store enriches local demand heatmaps and cohort mission quality. No competitor (PhonePe, Google Pay, BharatPe) can duplicate this without Paytm's vast network of 4.9 crore merchants. A network moat that deepens daily.



Merchant Lending Funnel

Cash Flow Prophet acts as real-time predictive underwriting. Stores with stable liquidity and growing cohort ranks become pre-qualified borrowers, feeding Paytm's highest-margin lending division at near-zero customer acquisition cost.

IMPLEMENTATION ROADMAP

1

Phase 1: Prove It (Sept 2026)

Working prototype with simulated data. Trade Radar + Festival Engine + Cash Flow Prophet. Sarvam voice briefings, Cognee memory graph, and n8n orchestration.

2

Phase 2: Pilot (3–6 Months)

Deploy to 500 merchants across Delhi NCR, Bengaluru, and Jaipur. Live Paytm transaction API hooks. Benchmark adoption, basket size uplift, and merchant engagement.

3

Phase 3: Scale (6–12 Months)

Roll out across 1.5 crore Soundbox subscribers. Add Price Pulse & collaborative Growth Missions. Expand to 11 Sarvam languages and sync with Paytm Lending.

Every Dukaan Deserves a Netrā

"Because the smartest businesses are not the biggest ones. They are the ones that can see the farthest."



TEAM KERNEL

Utkarsh Singh (Team Leader) & Pranav Pant

Track 1 — Merchant Growth AI · Paytm Build for India AI Hackathon (Delhi Edition)